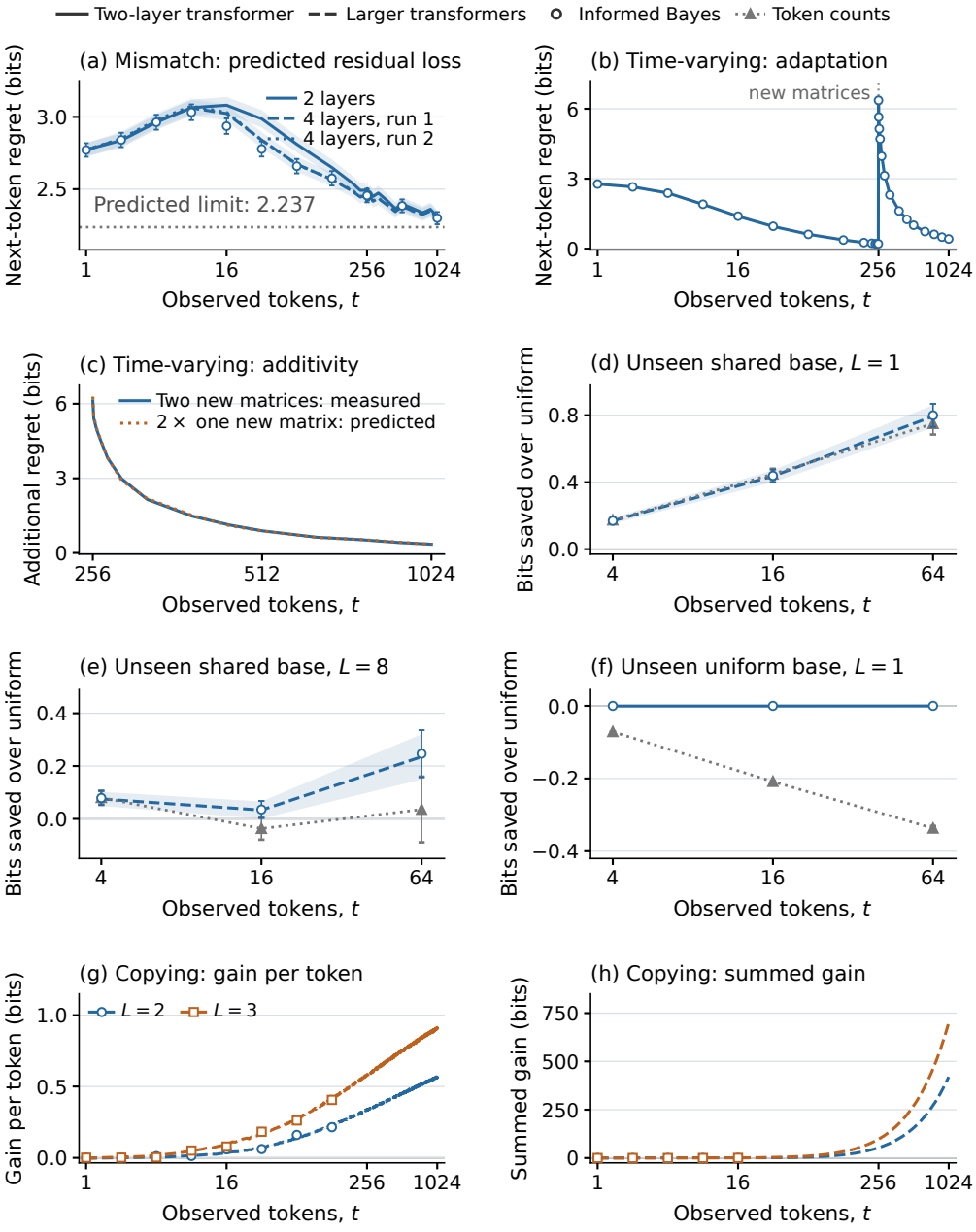

Two-layer transformer
Larger transformers
Informed Bayes
Token counts
(a) Mismatch: predicted residual loss
Next-token regret (bits)
2 layers
4 layers, run 1
4 layers, run 2
Predicted limit: 2.237
Observed tokens, t
(b) Time-varying: adaptation
new matrices
(c) Time-varying: additivity
Additional regret (bits)
Two new matrices: measured
2 × one new matrix: predicted
(d) Unseen shared base, L = 1
Bits saved over uniform
(e) Unseen shared base, L = 8
(f) Unseen uniform base, L = 1
(g) Copying: gain per token
Gain per token (bits)
L = 2
L = 3
(h) Copying: summed gain
Summed gain (bits)